Saketh Ayyagari

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Education

University of Maryland, College Park — College Park, MD
Bachelor of Science: Computer Science, Minor in Robotics | GPA: 4.0 — *May 2029*

Experience

Machine Learning Engineer — Feb 2026 - May 2026
Children's National Hospital — *College Park, MD*
- Developed a mobile application to detect safe vs. unsafe football tackles using pose estimation algorithms to support patient safety and research operations.
- Designed and implemented video recording and analysis features with 5 full-stack engineers, presenting progress in monthly design reviews with hospital representatives.
- Analyzed 70+ videos of football plays using Roboflow to train and evaluate pose estimation models tracking player joints.

Robotics Software Developer — Sep. 2025 - Present
Robotics at Maryland — *College Park, MD*
- Collaborated with mechanical, electrical, and software subteams to develop a Mars Rover for Canadian International Rover Challenge.
- Increased autonomous capabilities with depth cameras, IMU, and GPS into Gazebo to enhance localization.
- Researching feasibility of autonomous navigation pipelines using Nav2 and ArucoTags.

Software Engineering Intern — Sep. 2024 - Jan. 2025
Commvault — *Tinton Falls, NJ*
- Developed Python tool to visualize file metadata changes across 100+ files over a 7-day interval.
- Implemented anomaly detection functionalities using existing C++ code, file metadata, and SQL queries.
- Evaluated anomaly detection heuristics across file metadata changes to reduce false positives and improve detection reliability.

Research Intern — Jul. 2024 - Aug. 2024
MIT Beaver Works Summer Institute — *Cambridge, MA*
- Selected as 1 of 30 students for the Autonomous RACECAR Project.
- Developed computer vision, sensor fusion, and control algorithms with ROS2 to navigate a scale-model car; placed 3rd out of 7 teams in Grand Prix.
- Published research paper on IMU-based tire blowout detection; accepted to 2025 IEEE Vehicular Technology Conference.

Projects

Embodied AI Navigation Platform — Jun. 2026 – Present
- Developed visual SLAM pipeline in ROS2 using RTABMap to generate point cloud data from RGB-D cameras.
- Integrated YOLO image segmentation models to attach semantic understanding to 3D maps.
- Designing reusable ROS 2 packages and launch infrastructure to accelerate development of embodied AI navigation algorithms.

Simultaneous Localization and Mapping System — Feb. 2025 - Jun. 2025
- Fused LiDAR and odometry data from a robot for accurate localization and map generation of unknown environments using slam-toolbox.
- Integrated encoders, IMU, 2D LiDAR and Raspberry Pi 4B into a physical differential drive robot.
- Designed motor/sensor interfaces and teleoperation pipelines in ROS2 Python for real-time robot control and data logging.

Predicting Movement Direction with Machine Learning — Mar 2024 - Jan 2025
- Trained SVMs and Random Forests using scikit-learn, and Neural Networks with Tensorflow on neural spike data to predict direction of a mammal's movement.
- Tested 4 models on dataset of 143 neurons, achieving 75% peak accuracy with Random Forest.
- Drafted a 7-page research paper outlining methodology and results.

Skills and Certifications

Certifications: Claude Code 101
Languages: Python, Java, C/C++, PostgreSQL, HTML, JavaScript, Bash, MATLAB
Frameworks/Libraries: Tensorflow, PyTorch, NumPy, Scikit-learn, Matplotlib, ROS2, Gazebo, OpenCV, Streamlit
Tools: Git, GitHub, Docker, VS Code, Visual Studio, PyCharm, Eclipse, Roboflow, Microsoft Office, Google Suite